JUNCHENG LU

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EDUCATION

Teachers College, Columbia University — September 2025 – Present
M.A. in Cognitive Science in Education
GPA: 3.90 / 4.00

Beijing Normal University — September 2020 – July 2024
B.S. in Psychology
Selected coursework: Computational Cognitive Neuroscience; Brain Imaging Data Modeling; fMRI: Principles, Experiment Design and Data Analysis.

RESEARCH EXPERIENCE

Social Cognitive and Neural Sciences Lab — September 2025 – Present
Research Assistant, Columbia University | PI: Dr. Jon Freeman

Metacognitive Accuracy in Implicit Bias Prediction
First author; manuscript in preparation — January 2026 – Present

- Using Computational modeling in R to analyze 2,986 observations from 1,866 participants across four race and age Implicit Association Tests (IATs),
- Using maximum likelihood estimation to fit Multinomial Processing Tree (MPT)/Quadruple Process Model (Quad) to categorize participants' implicit biases into ingroup-affirming and outgroup-affirming specifications.
- Using Monte Carlo Markov Chain (MCMC) to fit hierarchical drift-diffusion model (HDDM) and hierarchical Quad model to detect latent processes of participants' awareness of their implicit biases.

Computational Modeling of the Weapon Identification Task
XXXX — September – December 2025

- Applied HDDM and MPT models to examine how evidence accumulation and latent processing components contribute to biased weapon-identification decisions.
- Compared competing cognitive architectures and related model-derived parameters to fMRI activation to evaluate accounts of threat processing and cognitive control.

Intergroup Relation Neuroscience Lab — March 2022 – June 2025
Research Assistant, Beijing Normal University | PI: Dr. Xiaochun Han

Computational and Neural Mechanisms of Collateral-Harm Decisions
R&R at Journal of Experimental Social Psychology; co-first author — March 2022 – June 2025

- Led study design, experimental implementation, and data collection in collaboration with the PI across two studies (total $N = 983$), combining a cross-cultural behavioral experiment (Chinese and U.S. participants) with EEG.
- Conducted all behavioral, computational, and EEG analyses, using HDDMs and EEGLAB in MATLAB to examine how civilian race and pain expressions activate neural responses related to decision caution.
- Identified a Race × Expression interaction in neural responses related to decision caution despite no corresponding condition differences in overt attack decisions. Contributed to manuscript drafting and revision with the PI.

Social Roles and Civilian–Soldier Judgments in Intergroup Conflict
Co-first author; manuscript in preparation — September 2023 – June 2025

- Conducted data analyses for four studies examining social-role information, civilian–soldier differences in attack decisions and bystander judgments, and representations of warmth and competence social impressions.
- Fit logistic and linear mixed-effects models accounting for participants and social roles to test cross-cultural patterns, category-by-information interactions, and experimental manipulations of warmth and competence.
- Analyses revealed smaller civilian–soldier differences with social-role information across decision and representation tasks. Contributed to manuscript writing.

MANUSCRIPTS

UNDER REVIEW

Lu, J.*, Zhang, Y.*, Han, X., Xiao, Z., & Han, X.† (2026). Computational and neural mechanisms of race-dependent influence of civilian suffering on collateral-harm decisions during intergroup conflicts. R&R at *Journal of Experimental Social Psychology*.

IN PREPARATION

Zhang, Y.*, **Lu, J.***, & Han, X.† (2026). Cognitive mechanism of killing-decision discrimination between combatants and civilians during intergroup conflict: The influence of social-role information. *Manuscript in preparation*.

* Equal contribution. † Corresponding author.

CONFERENCE

Lu, J., & Han, X.† (April 2024). Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. Accepted for oral and poster presentation at the *Social and Affective Neuroscience Society Annual Meeting*, Toronto, Canada. Unable to present due to visa restrictions.

TECHNICAL SKILLS

Programming	Python (HDDM, NumPy, pandas); R (brms, lme4, TreeBUGS, rtmpt); MATLAB (EEGLAB, SPM12, Psychtoolbox); JavaScript/HTML (jsPsych).
Modeling & Statistics	Hierarchical Bayesian modeling; DDM and MPT/Quad models; GLMM/LMM; MCMC diagnostics; model comparison.
Neuroimaging	EEG/ERP preprocessing and analysis; fMRI preprocessing and modeling; model-based brain–behavior analysis.